HW 2

Task 2) Why CTR bit flipping works w/o knowing the key

Formulas

Encryption: $C_i = M_i \oplus S_i$

Decryption: $M_i = C_i \oplus S_i$

'i' = 0x31 hex

'9' = 0x39 hex $\oplus$ = 0.08 hex

What happens during decryption. $M_i' = C_i \oplus S_i$

$= (C_i \oplus \Delta) \oplus S_i$

$= (M_i \oplus S_i \oplus \Delta) \oplus S_i$

$= M_i \oplus \Delta$

$M_i = 0x31 \oplus 0x08 = 0x39 = $ '9'

The attacker changes the plaintext w/o knowing the key because the same keystream is used during the decryption and cancels under XOR

Task 4) $M_i = C_i \oplus S_i$. $C_i = C_i \oplus X$

$M_i = C_i \oplus S_i$

$= (C_i \oplus X) \oplus S_i$

$(C_i \oplus S_i) \oplus X$

$= M_i \oplus X = 0x00$

$X = M_i$

XOR enables two different attacks. The original text does no null bytes so if it returns true after changing one space byte we know X = the original plaintext byte

Task 5)

of possible message is 256^n

b/c a byte has 256 values

so $(0-255)$

We know $M_i \oplus x = 0$

$x = M_i$

so each byte requires at most 255 queries

$$Q(n) \leq 255n$$

Each byte requires at most 255 oracle queries, so the total # of queries is $255n$ which grows linearly.

7) CBC decrypt: $M_i = D_k(c_i) \oplus c_{i-1}$

$c_{i-1} = c_{i-1} \oplus \Delta$

$M_i = D_k(c_i) \oplus c_{i-1}$

$= D_k(c_i) \oplus (c_{i-1} \oplus \Delta)$

$= (D_k(c_i) \oplus c_{i-1}) \oplus \Delta$

$= M_i \oplus \Delta$

Mod c_{i-1} by Δ mods the corresponding text block M_i by the same Δ w/o knowing the key.

8) attack block c_i needs a TWO Block Ciphertext

$C' = c_{i-1} \| c_i$

$M_i = D_k(c_i) \oplus IV$

$D_k(c_i) \oplus c_{i-1}$

$= M_i$

By Modding c_{i-1} we can attack any chosen plain text block.

We can take any two neighboring ciphertext blocks and decrypt them. This lets us use the padding oracle to test and eventually recover the plaintext without knowing the key!